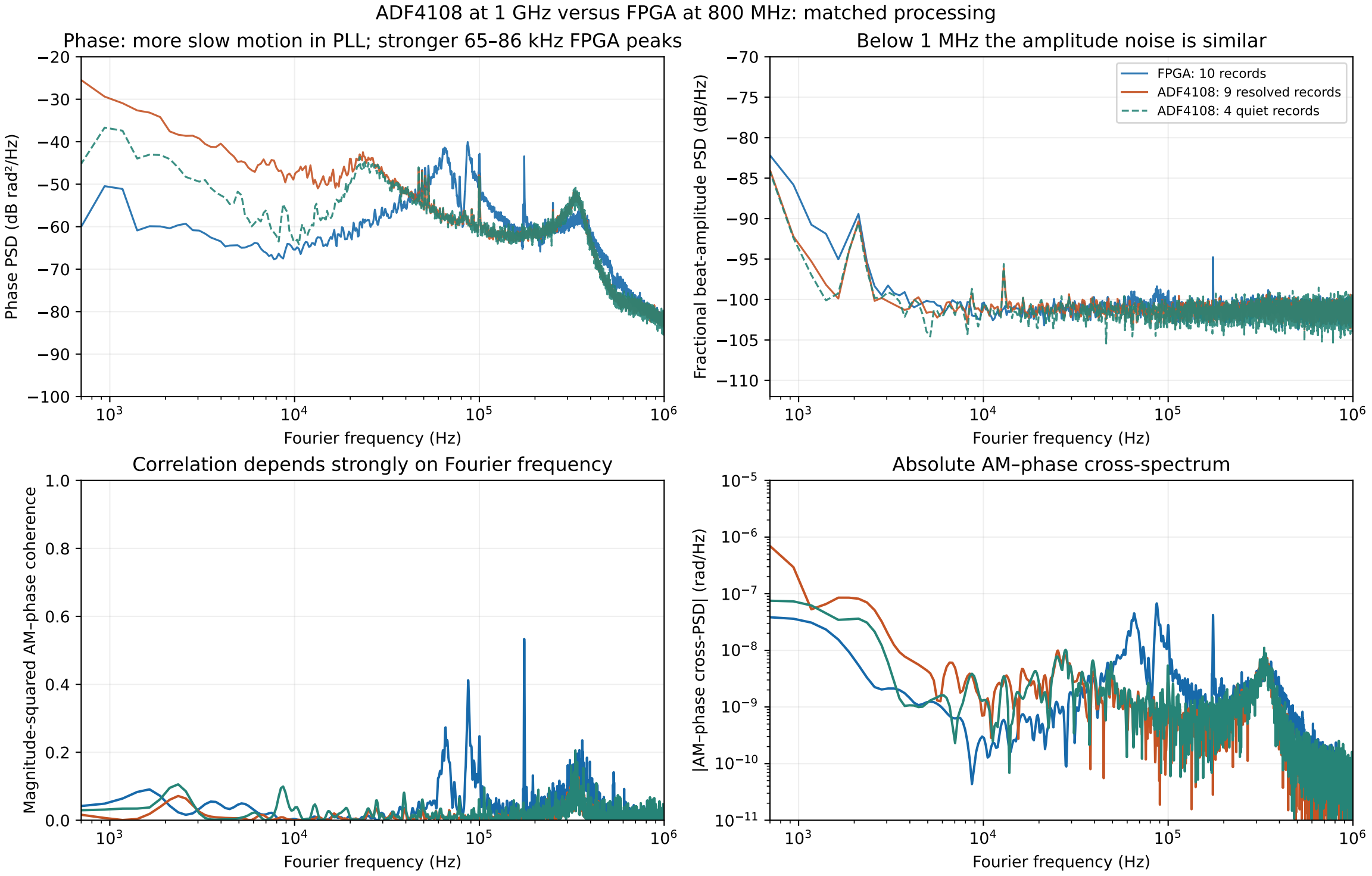

ADF4108 at 1 GHz versus FPGA at 800 MHz: matched processing
Phase: more slow motion in PLL; stronger 65–86 kHz FPGA peaks
Phase PSD (dB rad²/Hz)
Fourier frequency (Hz)
Below 1 MHz the amplitude noise is similar
Fractional beat-amplitude PSD (dB/Hz)
FPGA: 10 records
ADF4108: 9 resolved records
ADF4108: 4 quiet records
Fourier frequency (Hz)
Correlation depends strongly on Fourier frequency
Magnitude-squared AM–phase coherence
Fourier frequency (Hz)
Absolute AM–phase cross-spectrum
|AM–phase cross-PSD| (rad/Hz)
Fourier frequency (Hz)
Both: 15.36 MS/s, 17.07 ms records; 2 MHz complex low-pass removes the PLL's 3.2 MHz mixer image.
Spectra: 234.375 Hz bins; lower panels smoothed over ~1.1 kHz FWHM. PLL record 1 excluded: unwrap ambiguous.